

Part 0: IIT Delhi's Real BGP Connectivity

COL7524 – Advanced Computer Networks

Objective. This study investigates IIT Delhi's public BGP connectivity using two independent public BGP data sources: RIPEstat (RIPE RIS) and the Hurricane Electric BGP Toolkit. The investigated IPv4 prefix is 103.27.9.0/24, originated by AS132780 (Indian Institute of Technology Delhi).

1. Identification of IIT Delhi ASN and Prefix

Before tracking the active routing pathways, the network footprint of the target institution was manually identified.

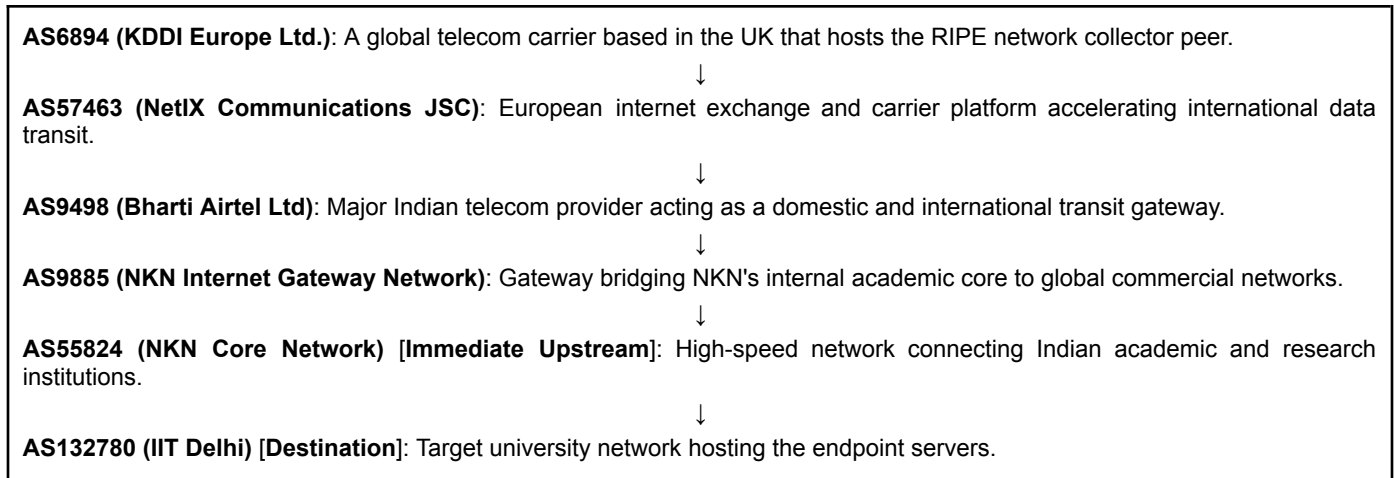
DNS Resolution: A manual DNS lookup was performed via the system command-line interface using `nslookup iitd.ac.in`. It resolved the domain name to the **IPv4 address: 103.27.9.24** (and the **IPv6 address: 2001:df4:e000:29::212**).

WHOIS Data Retrieval: A lookup for 103.27.9.24 shows that the covering route is 103.27.9.0/24, which has origin AS132780. Therefore, the **institute's public ASN is AS132780**, and **103.27.9.0/24** was selected for the BGP path analysis.

2. RIPEstat Observation

The current IPv4 prefixes for AS132780 are 103.27.8.0/24, 103.27.9.0/24, 103.27.10.0/24, and 103.27.11.0/24, as shown in the Announced Prefixes view in RIPEstat. The selected prefix 103.27.9.0/24 originates from AS132780 (IITDEL-AS-IN).

Observed RIPEstat path:



BGP Updates from 2026-08-13T10:00 UTC to 2026-08-13T11:00 UTC										
10	per page	Search...								
Prefix	Type	Path							Communities	Peer
103.27.9.0/24	A	AS6894	AS57463	AS9498	AS9885	AS9885	AS9885	AS55824	AS132780	5.57.81.186 01
103.27.9.0/24	A	AS6894	AS57463	AS9498	AS9885	AS9885	AS9885	AS55824	AS132780	5.57.81.186 01

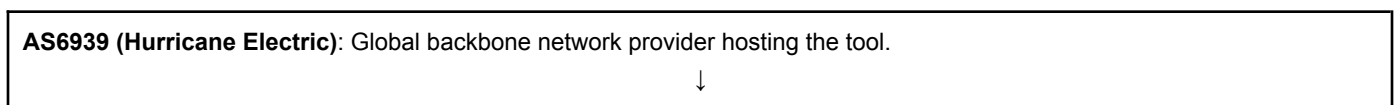
Showing 1 to 2 of 2 entries

Figure 1. RIPEstat BGP update showing the observed path for 103.27.9.0/24.

3. Hurricane Electric Observation

The Hurricane Electric BGP Toolkit also identifies AS132780 as the origin for 103.27.9.0/24.

Observed Hurricane Electric path:



AS6453 (Tata Communications): An international Tier-1 carrier that manages global routing lanes.



AS4755 (Tata India): Domestic telecom network routing traffic toward NKN.



AS9885 (NKN Gateway): Boundary network bridging commercial providers to NKN's core.



AS55824 (NKN Core) [Immediate Upstream]: Academic network backbone directly connected to IIT Delhi.



AS132780 (IIT Delhi) [Destination]: Target university network hosting the endpoint servers.

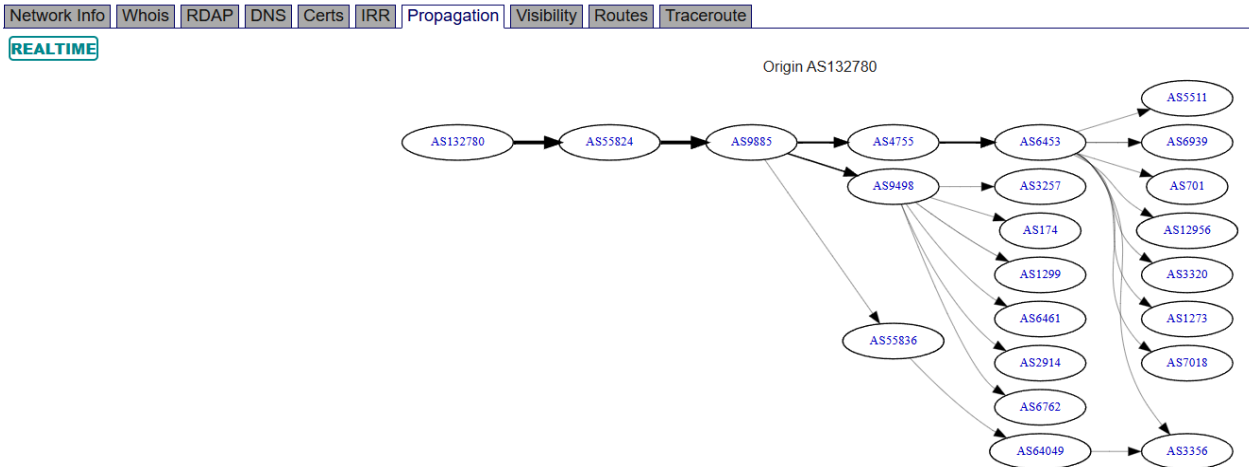


Figure 2. Hurricane Electric BGP Toolkit topology for origin AS132780.

4. Upstream Connectivity Hierarchy

The observations establish the immediate connectivity relationship: AS132780 (IIT Delhi) → AS55824 (NKN Core Network). Following the RIPEstat path beyond NKN gives four more AS hops: AS9885 → AS9498 → AS57463 → AS6894. In contrast, the Hurricane Electric observation shows a different upstream continuation after AS9885: AS4755 → AS6453 → AS6939.

5. Cross-Tool Comparison

Analysis Metric	RIPEstat	Hurricane Electric
Investigated prefix	103.27.9.0/24	103.27.9.0/24
Origin ASN	AS132780 (IIT Delhi)	AS132780 (IIT Delhi)
Immediate upstream	AS55824 (NKN Core)	AS55824 (NKN Core)
Observed path	AS6894 → AS57463 → AS9498 → AS9885 → AS55824 → AS132780	AS6939 → AS6453 → AS4755 → AS9885 → AS55824 → AS132780

6. Comparisons and Conclusion

Both tools agree on the origin of the investigated prefix, AS132780, and on the immediate AS adjacent to IIT Delhi, AS55824 (NKN Core Network). The difference appears in the observed upstream continuation beyond NKN. This behaviour is expected, as BGP is distributed: different route collectors and vantage points can see different valid paths to the same prefix. The two paths should be seen as observations from different points on the Internet, not as contradictions.

Overall, the results show IIT Delhi operating AS132780 and using the NKN Core Network as its immediate upstream. In contrast, the broader upstream connectivity visible from the Internet depends on the observation point.